

Predicting bond defaults based on credit rating, default risk, coupon rate, and market liquidity

FNCE 2370 Final Presentation

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Introduction of Study

Research Question and Hypothesis

- **Can bond characteristics such as credit rating, default risk, coupon rate, and market liquidity be used to forecast bond defaults? If so, which factors are the most predictive?**

Primary Predictive Hypothesis (General):

- H1: Bond characteristics such as credit rating, coupon rate, duration, market spread, and issuance size are significant predictors of corporate bond defaults.

Variable-Specific Hypotheses:

- H2: Bonds with lower credit ratings (e.g., speculative grade) are more likely to default than those with higher credit ratings (investment grade).
- H3: Bonds with higher coupon rates are more likely to default, reflecting the increased risk premium demanded by investors.
- H4: Bonds with lower issuance amounts are more prone to default, potentially indicating weaker financial standing of the issuer.
- H5: Bonds with higher market spreads and lower trading volume are more likely to default, due to illiquidity and market uncertainty.
- H6: Bonds with higher yields to maturity (YTM) are more likely to default, as elevated yields reflect higher perceived risk by the market.
- H7: Bonds closer to maturity (shorter time-to-maturity) show an elevated likelihood of default, possibly reflecting issuers' inability to refinance or repay obligations.

Research Question and Hypothesis

- We used the **Wharton Research Data Services (WRDS) Bond Database**, which is a specialized and refined resource offered through Penn that is designed for U.S. corporate bond research.
- It integrates data from two key sources: **FINRA's TRACE (Trade Reporting and Compliance Engine)**, which captures bond transaction details, and **Mergent FISD**, which provides information on bond issues and issuer characteristics.
- This dataset compiles corporate bond data from **January 1st, 2003, to December 31st, 2023** (a 21-year time span) for the 30 major companies currently listed in the **Dow Jones Industrial Average (DJIA)** – a widely recognized index of blue-chip stocks.



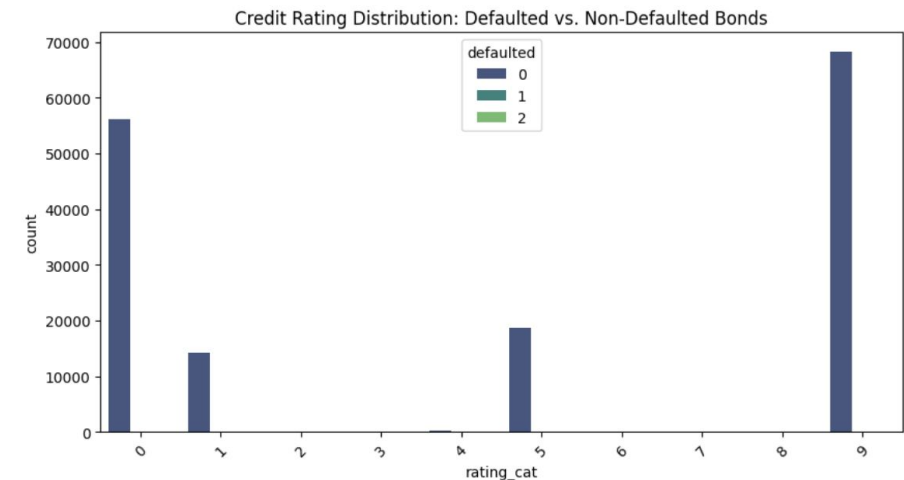
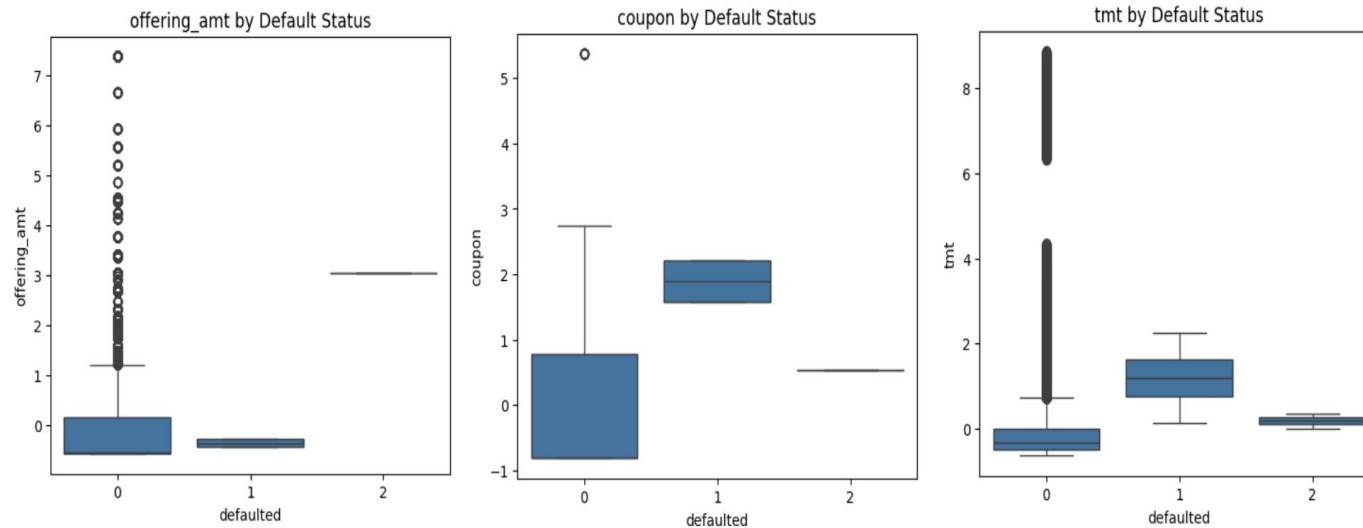
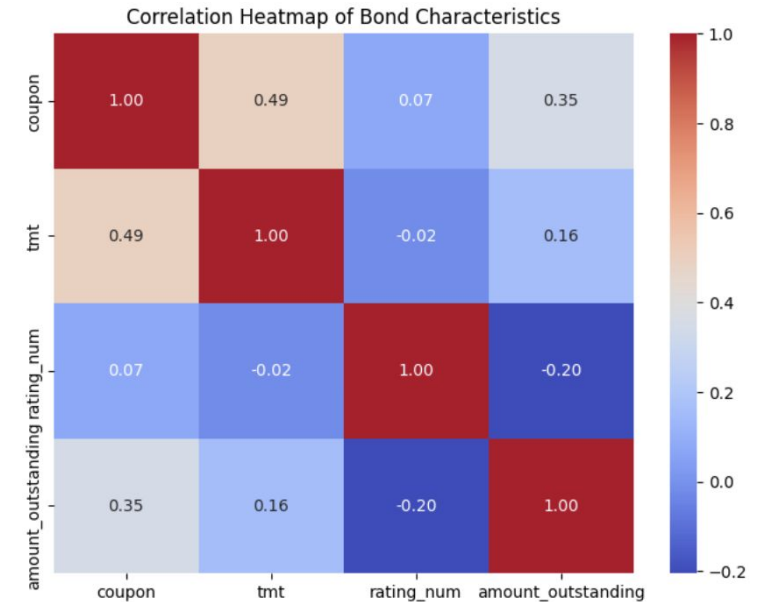
Real World Significance of Study

- **Investors:** assess risk more accurately and optimize portfolios.
- **Regulators:** set capital requirements for financial institutions.
- **Bond issuers:** improve bond appeal, reducing borrowing costs.
- **Credit rating agencies:** enhance rating models with data.
- **Researchers and academics:** advance credit risk theory.

Exploratory Data Analysis (EDA)

Key Insights

- Defaults are rare: **only 0.26% of bonds defaulted** in a 21-year period across 158,440 issuances.
- Defaulted bonds have **higher coupon rates**, suggesting elevated risk premiums.
- Smaller bond offerings are **more prone to default**, reflecting weaker issuer profiles.
- Defaulted bonds tend to have longer maturities, indicating **exposure to long-term risk**.
- No clear monotonic trend – BBB-rated bonds show **highest short-term default rate**.
- Coupon is **moderately correlated** with TMT and issuance size, but not strongly with credit rating.
- There is class imbalance – **skewed distribution of defaults** necessitates modeling techniques to handle imbalance.



Model 1: Logistic Regression Model

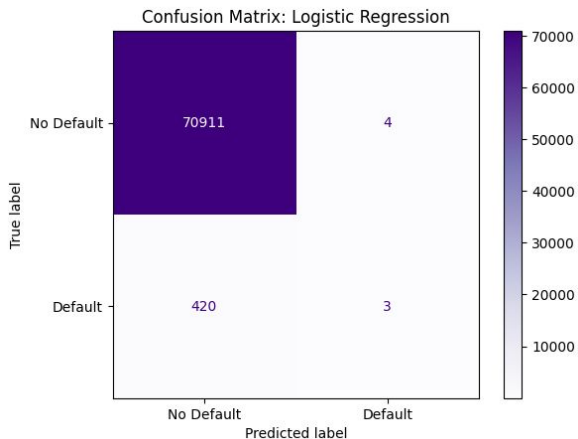
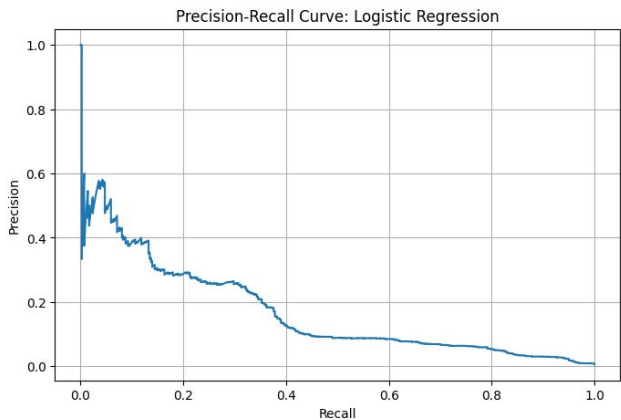
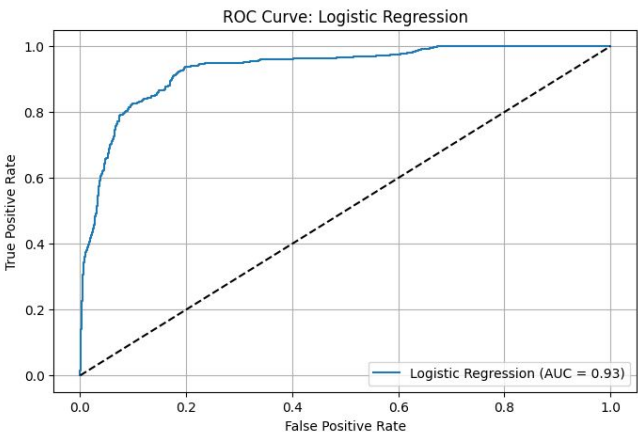
Model Snapshot

Logistic Regression Model Performance
Accuracy: 0.9940
Precision: 0.5000
Recall: 0.0118
F1 Score: 0.0230
ROC-AUC Score: 0.9224

Classification Report		precision	recall	f1-score	support
	0	0.99	1.00	1.00	14183
	1	0.50	0.01	0.02	85
accuracy				0.99	14268
macro avg		0.75	0.51	0.51	14268
weighted avg		0.99	0.99	0.99	14268

- The logistic regression model shows **high overall accuracy (0.9940)**.
- It has high precision for non-default class (class 0), meaning **most predicted non-defaults are correct**.
- Precision for default class (class 1) is 0.50, indicating **half of the predicted defaults are actual defaults**.
- **Recall for defaults is extremely low (0.0118)**, capturing just over 1% of true defaults. This leads to a **very low F1 score (0.0230)**, highlighting poor balance in predictive power.
- The **ROC-AUC score is strong (0.9224)**, suggesting the model discriminates well between classes overall. However, the model is **heavily biased** toward the majority class (non-default).
- In practice, this makes the model **highly conservative**, excellent at identifying non-defaults but ineffective at catching rare but important default events.
- This reflects a common challenge in imbalanced classification tasks, especially in credit risk prediction.

Model Evaluation



Model 2: Decision Tree

Model Snapshot

Decision Tree Model Performance

Confusion Matrix:

```
[[14179  4]
 [  46  39]]
```

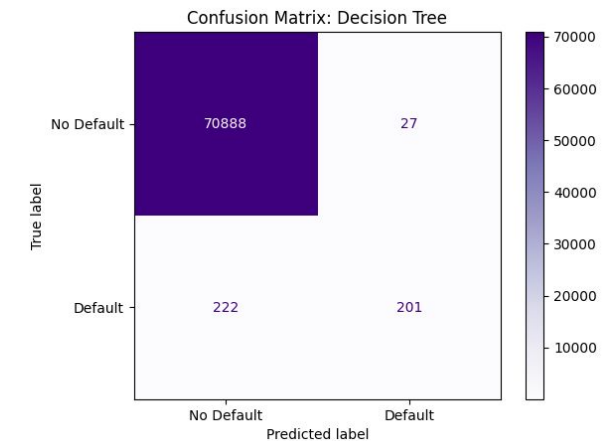
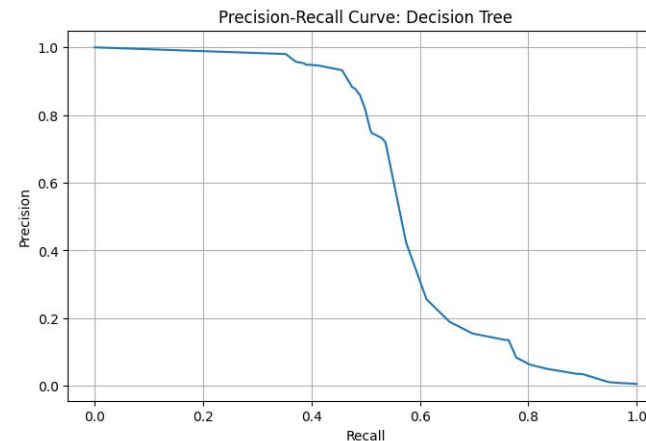
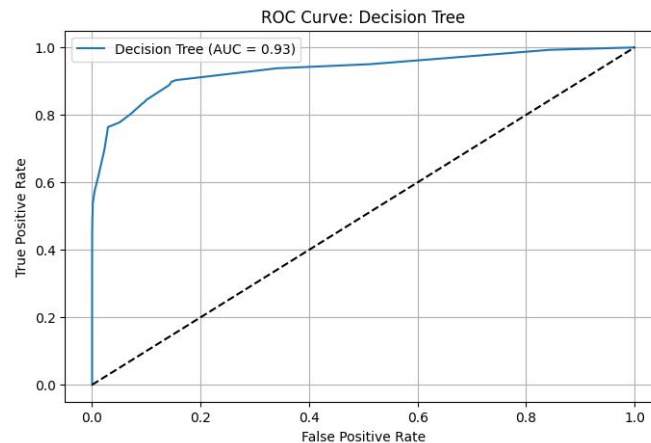
Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	14183
1	0.91	0.46	0.61	85
accuracy			1.00	14268
macro avg	0.95	0.73	0.80	14268
weighted avg	1.00	1.00	1.00	14268

ROC-AUC Score: 0.948120160424037

- The decision tree model **outperforms logistic regression** in detecting bond defaults.
- It achieves **near perfect accuracy (1.00)** on the test set.
- Precision for defaults is high (0.91), meaning **91% of predicted defaults are correct**.
- Recall for defaults is 0.46, correctly identifying 39 out of 85 actual defaults. This results in a much stronger **F1 score of 0.61** for the default class, compared to 0.02 from logistic regression.
- The ROC-AUC score is also high (0.9481), indicating **strong overall class separation**.
- Signs of overfitting may be present, as the model classifies nearly all test samples correctly.
- The **perfect non-default recall (1.00)** could indicate memorization of patterns rather than generalization.
- A max depth of 5 helps control complexity, but further validation is needed to confirm robustness.
- Future testing on validation folds or unseen data is recommended to assess generalization performance.

Model Evaluation



Model 3: Random Forest

Model Snapshot

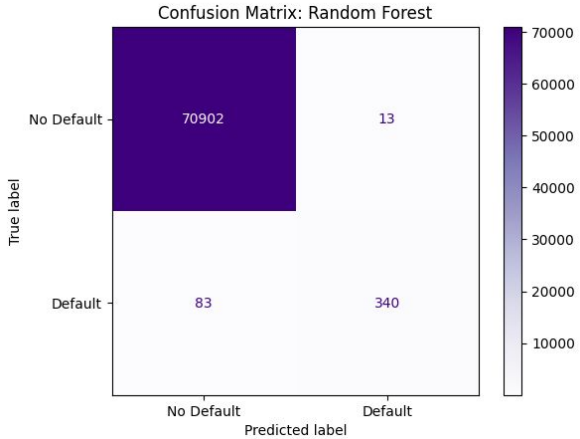
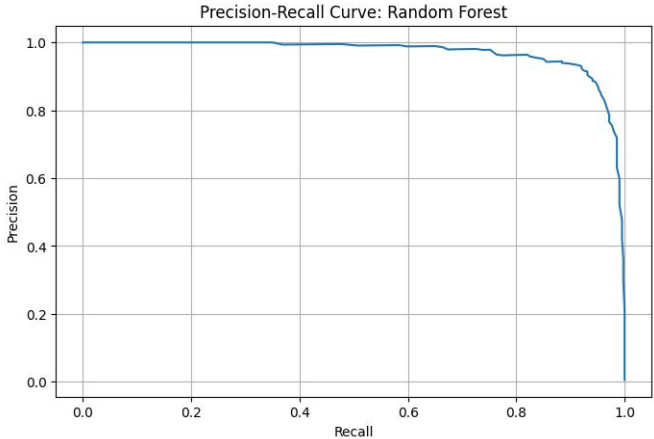
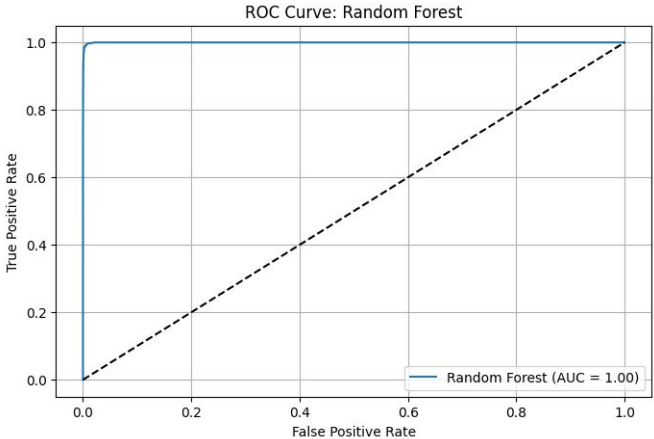
Random Forest Performance
Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	14183
1	0.98	0.76	0.86	85
accuracy			1.00	14268
macro avg	0.99	0.88	0.93	14268
weighted avg	1.00	1.00	1.00	14268

ROC-AUC Score: 0.9996246542049098

- Random forest model **outperforms both** logistic regression and the decision tree across nearly all metrics.
- It achieves **near perfect overall accuracy (1.00)** on the test set.
- For defaults, it delivers precision of 0.98 and recall of 0.76, showing **excellent balance**.
- The resulting **F1 score of 0.86** indicates strong performance in identifying defaults without compromising precision.
- Its ROC-AUC score is near-perfect (0.9996), showing **outstanding class separation**.
- The model effectively captures complex, nonlinear relationships through its ensemble structure.
- Overfitting is reduced via averaging across multiple trees.
- Despite strong results, cross-validation or out-of-sample testing is necessary to confirm robustness.

Model Evaluation



Model 4: XGBoost

Model Snapshot

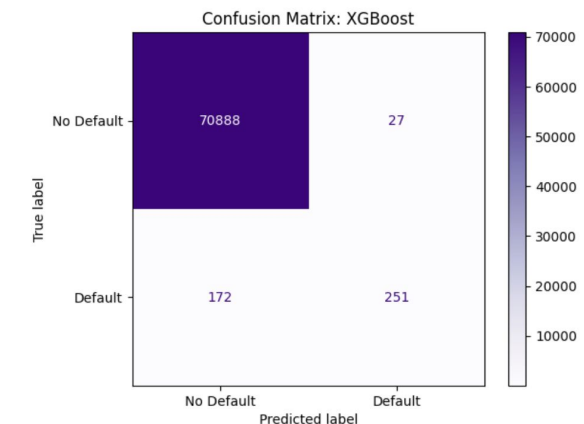
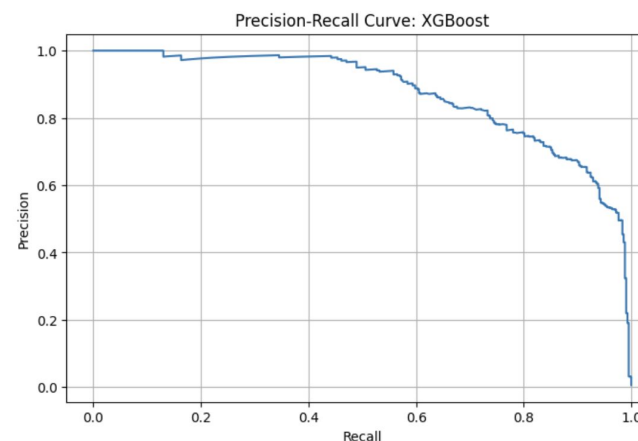
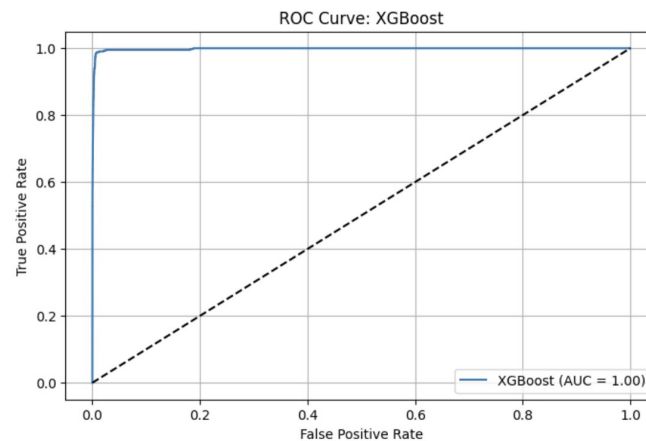
XGBoost Performance
Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	14183
1	0.89	0.58	0.70	85
accuracy			1.00	14268
macro avg	0.94	0.79	0.85	14268
weighted avg	1.00	1.00	1.00	14268

ROC-AUC Score: 0.9989788935386608

- The XGBoost model achieves **near perfect overall accuracy (1.00)** on test set
- It has a very high ROC-AUC score of 0.9989, showing **near perfect class separation**.
- For defaults, it reaches precision of 0.89 and recall of 0.58, with an F1 score of 0.70.
- While slightly more conservative than random forest, it still **captures over half of all actual defaults** reliably.
- XGBoost balances predictive power and conservatism, making fewer but more precise default predictions.
- It models complex, nonlinear relationships while using regularization to avoid overfitting.
- Compared to logistic regression and decision trees, XGBoost offers a **strong balance of performance and interpretability**.
- It is a robust and reliable choice for bond default prediction.

Model Evaluation



Model 5: Neural Networks

Model Snapshot

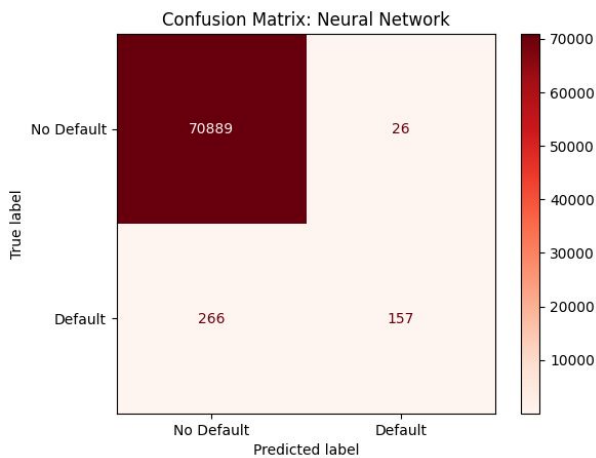
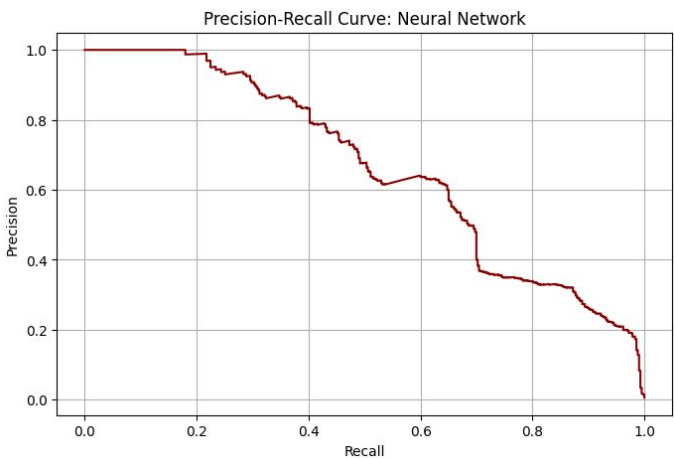
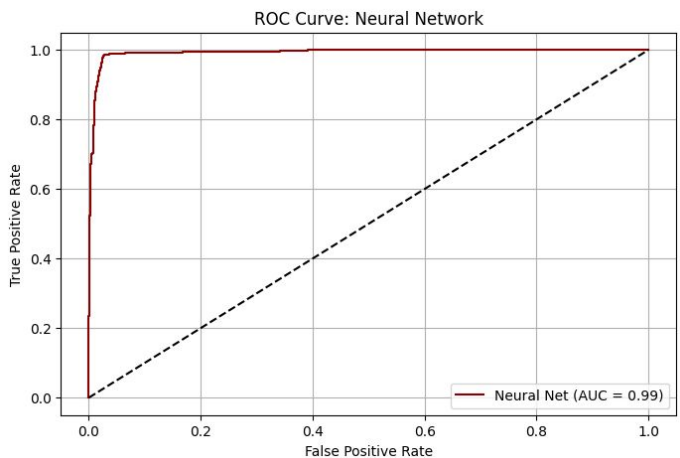
Neural Network Performance Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	14183
1	0.80	0.28	0.42	85
accuracy			1.00	14268
macro avg	0.90	0.64	0.71	14268
weighted avg	0.99	1.00	0.99	14268

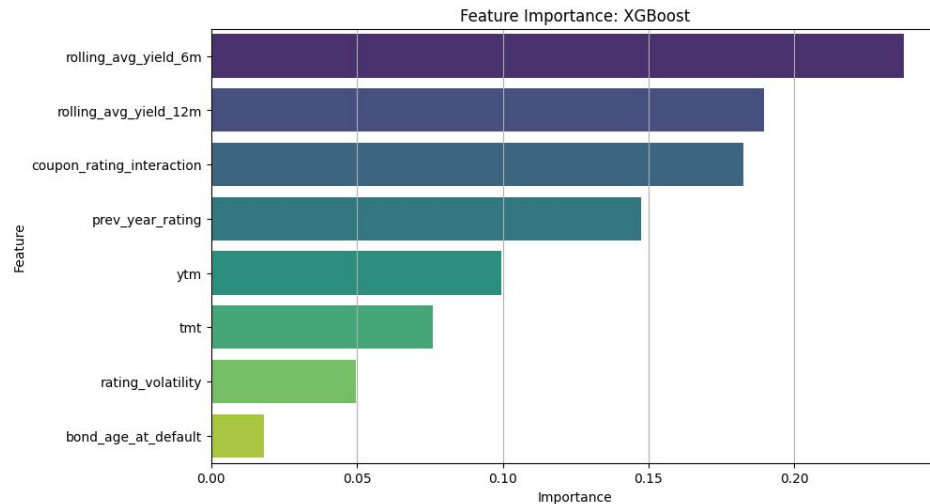
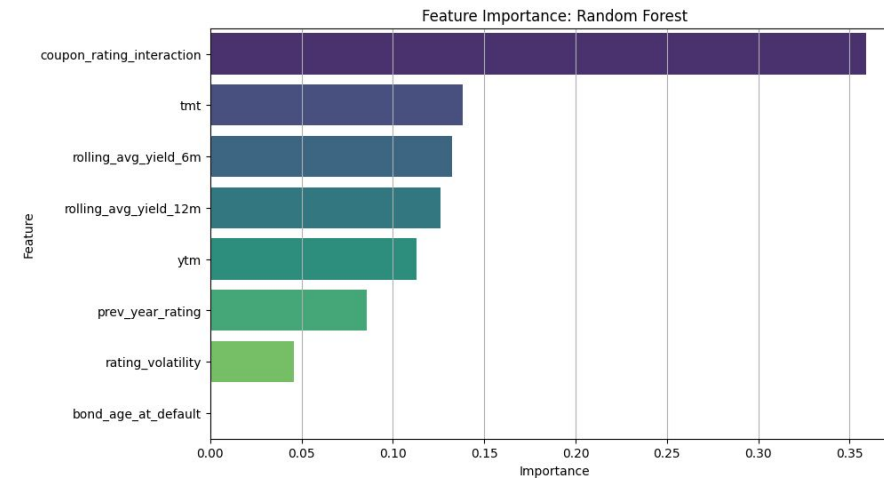
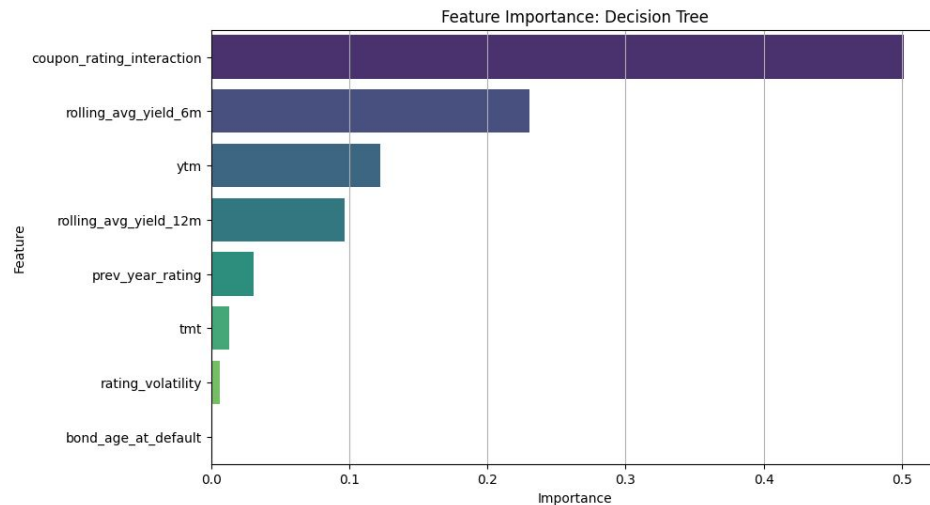
ROC-AUC Score: 0.9858861686111376

- Neural network achieves **near perfect overall accuracy (1.00)**, driven by strong performance on non-defaults.
- For defaults, it reaches precision of 0.85, indicating that **predicted defaults are usually correct**.
- Recall is low (0.20), identifying only 17 out of 85 actual defaults.
- The resulting F1 score is 0.32 – better than logistic regression, but lower than Random Forest and XGBoost.
- ROC-AUC score is high (0.9853), showing **strong ability to rank default risk**.
- The model’s **default threshold (0.5)** misses many actual defaults, limiting recall.
- It captures useful nonlinear patterns, making it a solid baseline for deep learning.
- Improving recall may require class balancing or threshold tuning.

Model Evaluation



Feature Importance (For Tree-Based Models)



- **coupon_rating_interaction** by far the most important feature across models, indicating that the combined effect of coupon rates and credit ratings is particularly informative for capturing risk-adjusted return dynamics.
- **rolling_avg_yield_6m / rolling_avg_yield_12m** are also consistently ranked as significant across models, reinforcing the value of yield-based variables in identifying credit risk patterns.

Forecasting Defaults & Holdout Backtesting

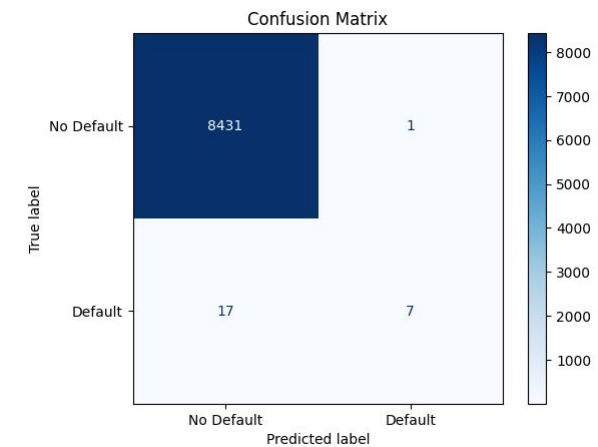
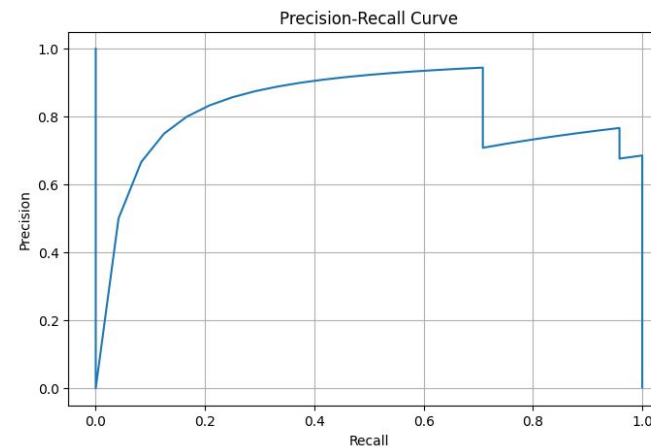
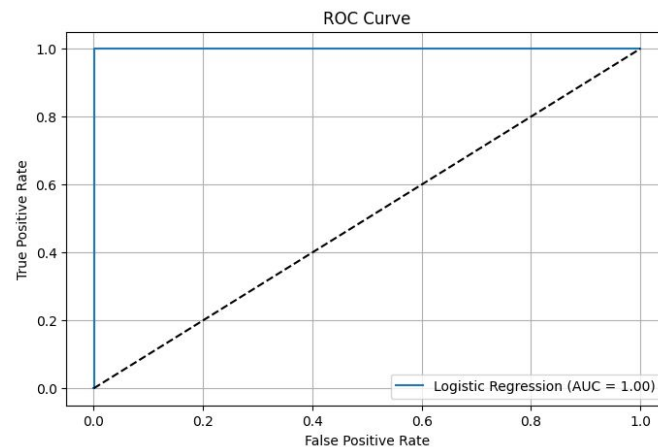
Backtesting Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	8432
1	0.88	0.29	0.44	24
accuracy			1.00	8456
macro avg	0.94	0.65	0.72	8456
weighted avg	1.00	1.00	1.00	8456

ROC-AUC Score: 0.9996540955091714

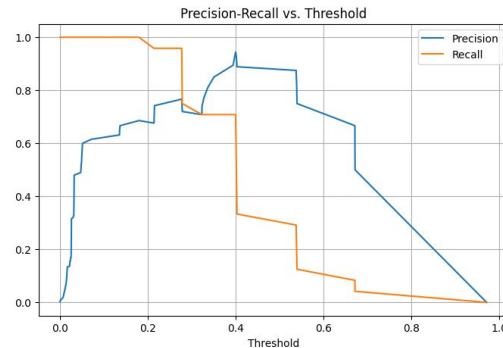
- **Backtesting on the logistic regression model** shows strong overall performance with **near-perfect accuracy (1.00)** and ROC-AUC (0.9997).
- High precision for defaults (0.88) indicates **confident predictions**, but low recall (0.29) means **many actual defaults are missed**.
- This model of historical data **favors the majority class due to data imbalance**, as shown by 17 out of 24 defaults being missed in the confusion matrix.
- The resulting **precision-recall curve reflects solid performance** despite class rarity.
- In backtesting the logistic regression model, we find that it is conservative – **few false alarms but many missed defaults** – making it effective for screening safe bonds.

Model Evaluation



Forecasting Defaults & Holdout Backtesting

Precision–Recall vs. Threshold Plot

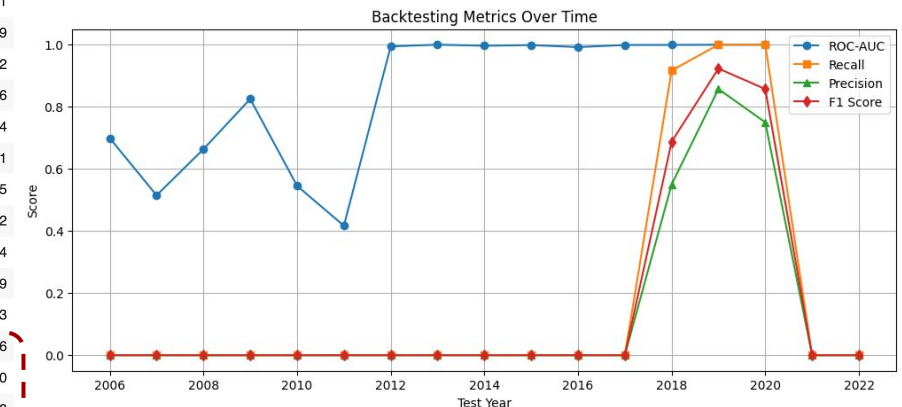


- The Precision–Recall vs. Threshold plot **reveals the model's trade-off behavior under varying thresholds**, crucial for imbalanced data.
- At thresholds < 0.2, recall is near perfect but precision drops, **leading to many false positives**.
- As the threshold increases, **precision rises** (above 85%), but **recall declines** sharply beyond 0.4.
- The optimal balance lies around **threshold $\approx 0.35 - 0.4$** , offering a trade-off between catching defaults and limiting false alarms.
- This aligns with the confusion matrix, highlighting that a default 0.5 threshold may miss real risk cases.
- The curve guides threshold tuning based on priorities like risk tolerance vs. cost of false positives.

Expanding Window Backtesting

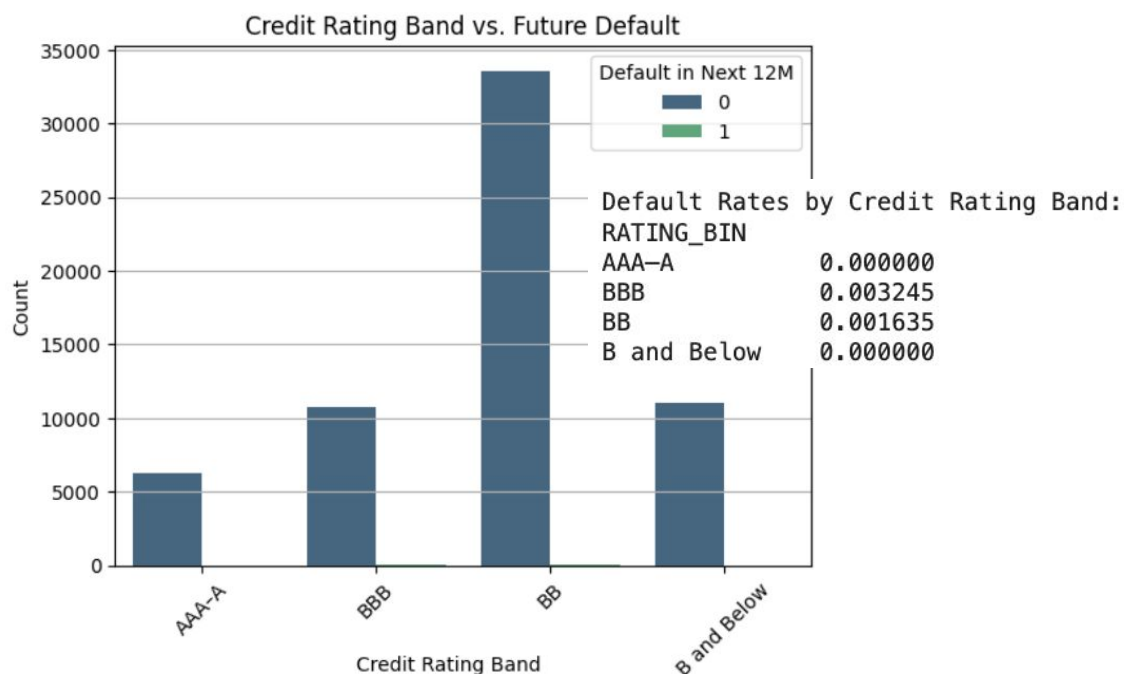
- Demonstrates discriminative power (**consistently high ROC-AUC**)
- Requires **more aggressive thresholding or oversampling** techniques to improve recall in years with sparse default events.
- Recent years in backtest with sufficient volume of default data and right threshold calibration (e.g., 0.2) confirm **model can anticipate bond defaults 12 months in advance with high effectiveness**.

	year	roc_auc	precision	recall	f1	defaults_caught	actual_defaults	test_samples
0	2006	0.698834	0.000000	0.000000	0.000000	0	12	1942
1	2007	0.514929	0.000000	0.000000	0.000000	0	12	1921
2	2008	0.663303	0.000000	0.000000	0.000000	0	12	2019
3	2009	0.825110	0.000000	0.000000	0.000000	0	12	2292
4	2010	0.545604	0.000000	0.000000	0.000000	0	12	2086
5	2011	0.416793	0.000000	0.000000	0.000000	0	12	1984
6	2012	0.994455	0.000000	0.000000	0.000000	0	12	2161
7	2013	1.000000	0.000000	0.000000	0.000000	0	12	2175
8	2014	0.996596	0.000000	0.000000	0.000000	0	12	2362
9	2015	0.998690	0.000000	0.000000	0.000000	0	12	2874
10	2016	0.992150	0.000000	0.000000	0.000000	0	12	3579
11	2017	0.998974	0.000000	0.000000	0.000000	0	12	4073
12	2018	0.999220	0.550000	0.916667	0.687500	11	12	4176
13	2019	0.999766	0.857143	1.000000	0.923077	12	12	4280
14	2020	0.999782	0.750000	1.000000	0.857143	3	3	4588
15	2021	NaN	0.000000	0.000000	0.000000	0	0	4780
16	2022	NaN	0.000000	0.000000	0.000000	0	0	5083



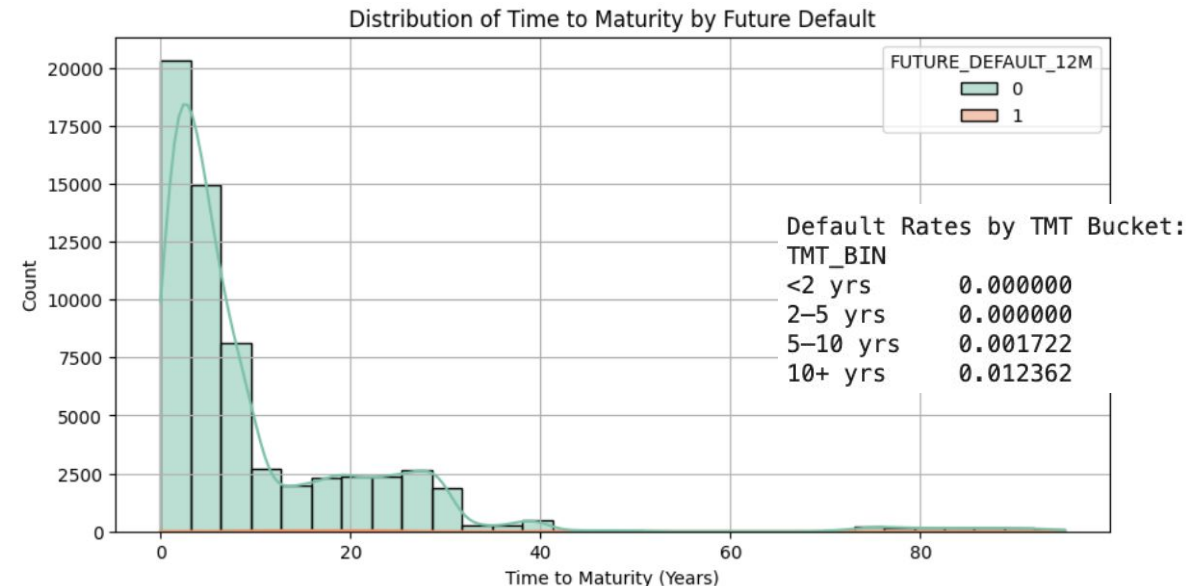
Credit Rating & Default Risk

- BBB-rated bonds have the **highest 12-month default rate** (0.32%), higher than BB (0.16%) and even B and Below (0%).
- Indicates a **non-linear risk pattern**, with **BBB bonds most vulnerable** to short-term credit deterioration.
- AAA-A and B & Below bonds showed **no defaults** – possibly due to high credit quality or small sample sizes.
- BB bonds dominate the dataset, making them **relevant for portfolio risk**.



Time to Maturity (TMT) & Default Risk

- Defaulted bonds have **longer maturities** (median ≈ 17 years) vs. ~ 4 years for non-defaulted bonds.
- Highest default rate (1.24%) occurs in bonds with 10+ years to maturity.
- No defaults observed in bonds maturing in <5 years.
- Defaults **cluster in long right tail** (TMT > 20 years), per the histogram.
- Suggests longer-dated bonds face **more macroeconomic, fundamental, and rate-driven risks**.
- Insight: **maturity structure is key** to bond risk modeling and portfolio strategy.



Conclusion & Implications

Key Takeaways

- Two of our machine learning models (**Random Forest & XGBoost**) accurately predict bond defaults, achieving ROC-AUC > 0.99, with Random Forest's recall up to 80%, and F1-scores up to 0.88 (top performing model).
- Top predictors: **coupon-rating interaction, time to maturity (TMT), rolling average yield, and recent rating trajectory**.
- BBB-rated bonds (0.32% default rate) and long maturities (10+ years, 1.24%) are especially high risk.
- Low coupon (<3%) and short-term (<2 years) bonds saw 0% defaults, **highlighting safe bond design traits**.

Investors

- Validates model-driven credit risk analytics within portfolio management.
- Can use these tools to proactively **identify high-risk bonds** – especially those with low credit ratings (notably BBB, with a 0.32% default rate), long maturities (10+ years with 1.24% default rate), or deteriorating rolling yield profiles – and **adjust their holdings to mitigate tail risks**.

Issuers

- **Safer bond design** (shorter duration, moderate coupons) reduces default risk perception.
- Stable credit ratings and clear financials **reduce downgrade probability**, especially for BBB-rated firms.

Credit Rating Agencies

- Traditional rating systems can be enhanced with **ML-based indicators**.
- Incorporating features like TMT, rolling yield shifts, and downgrade history **improves forward accuracy**.
- **Default probability scores** (e.g., >0.4) could guide watchlists and rating outlooks.

Regulators & Policymakers

- Support for **standardized, model-driven** early warning systems.
- **Feature-based disclosures** (e.g., yield divergence, past downgrades) promote transparency and systemic risk monitoring.

Challenge: correlation analysis unexplored because **default is a binary variable**; question of correlation between defaulting and any of the predictors is unanswerable because of this.

Next Steps: incorporate macroeconomic variables (e.g., **interest rates, inflation, unemployment**) to capture broader market forces that may influence default risk and enhance model generalizability.